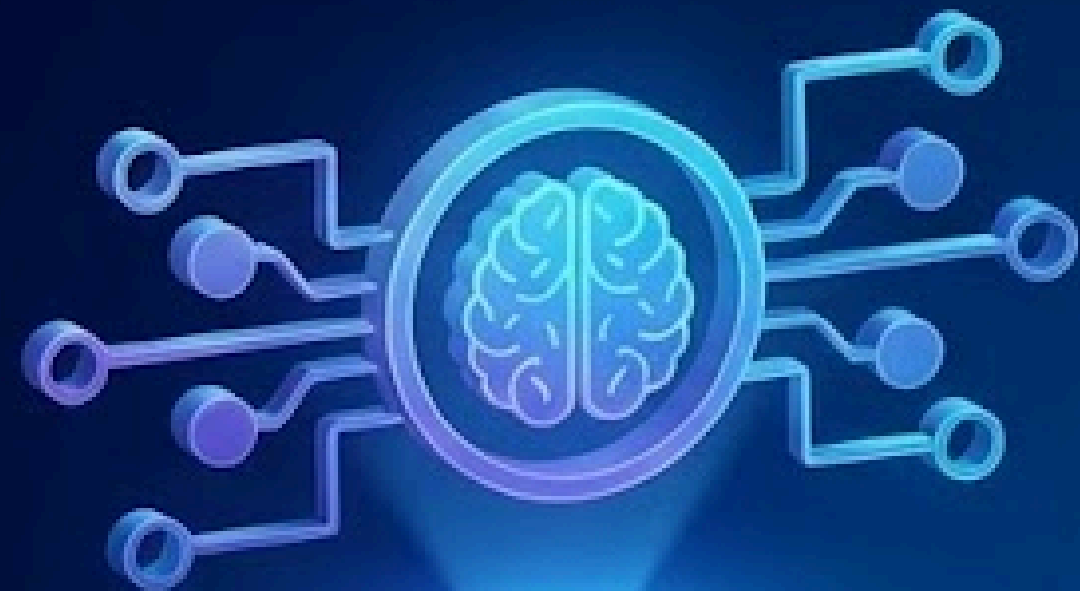




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Q.2 (a) Let $A = B = \{-1 \leq x \leq 1\}$. For each of the following functions, state whether it is injective, surjective, or bijective: [7]

i) $g(x) = \sin(\pi x)$

ii) $b(x) = \lfloor x \rfloor / 3$ (Note: Transcribed as floor function based on context)

(b) Let $f(x) = x + 2$, $g(x) = x - 2$, $h(x) = 3x$. Find: [7]

(i) $f \circ g$

(ii) $f \circ g \circ h$

Q.2 (a) Let $A = B = \{-1 \leq x \leq 1\}$. For each of the following functions, state whether it is injective, surjective, or bijective: [7]

i) $g(x) = \sin(\pi x)$

ii) $b(x) = \lfloor x \rfloor / 3$ (Note: Transcribed as floor function based on context)

Problem: Let $A = B = [-1, 1]$. State whether the following are injective, surjective, or bijective.

i) $g(x) = \sin(\pi x)$

- **Analysis:** The domain is $[-1, 1]$. The term πx ranges from $[-\pi, \pi]$.
- **Injectivity (One-to-One):** Let's check values. $g(0) = \sin(0) = 0$. $g(1) = \sin(\pi) = 0$. Since distinct inputs (0 and 1) map to the same output (0), the function is **NOT Injective**.
- **Surjectivity (Onto):** The range of $\sin(\theta)$ over $[-\pi, \pi]$ is $[-1, 1]$. The codomain B is $[-1, 1]$. Since Range = Codomain, the function is **Surjective**.
- **Bijectivity:** Since it is not injective, it is **NOT Bijective**.

ii) $b(x) = \lfloor x \rfloor / 3$

- **Analysis:** $\lfloor x \rfloor$ is the floor function (greatest integer $\leq x$). For $x \in [-1, 1]$, the possible integer values of $\lfloor x \rfloor$ are $-1, 0, 1$. Thus, the possible outputs are $\{-1/3, 0, 1/3\}$.
- **Injectivity:** Consider $x = 0.1$ and $x = 0.5$. $b(0.1) = \lfloor 0.1 \rfloor / 3 = 0/3 = 0$. $b(0.5) = \lfloor 0.5 \rfloor / 3 = 0/3 = 0$. Multiple inputs map to the same output. **NOT Injective.**
- **Surjectivity:** The codomain is the interval $[-1, 1]$. The range is only the discrete set $\{-1/3, 0, 1/3\}$. Since the range does not cover the entire interval, it is **NOT Surjective.**
- **Bijectivity: NOT Bijective.**

(b) Let $f(x) = x + 2$, $g(x) = x - 2$, $h(x) = 3x$. Find:

[7]

(i) $f \circ g$

(ii) $f \circ g \circ h$

(i) Find $f \circ g$

The notation $(f \circ g)(x)$ means $f(g(x))$.

$$(f \circ g)(x) = f(g(x))$$

$$= f(x - 2)$$

$$= (x - 2) + 2$$

$$= x$$

(Substitute $g(x)$)

(Apply f)

Result: $(f \circ g)(x) = x$ (Identity Function).

(ii) Find $f \circ g \circ h$

The notation $(f \circ g \circ h)(x)$ means $f(g(h(x)))$.

Step 1: Evaluate inner composition $g(h(x))$

$$\begin{aligned}g(h(x)) &= g(3x) \\ &= 3x - 2\end{aligned}$$

Step 2: Evaluate outer composition $f(\dots)$

$$\begin{aligned}f(g(h(x))) &= f(3x - 2) \\ &= (3x - 2) + 2 && \text{(Apply } f\text{)} \\ &= 3x\end{aligned}$$

Result: $(f \circ g \circ h)(x) = 3x$.

Q.3 (a) Find the power set of each of these sets: [7]

i) $\{a, b\}$

ii) $\{\phi, \{\phi\}\}$

(b) Use Cantor's diagonal argument to prove that the set F of all functions $f : (0, 1) \rightarrow \mathbb{R}$ has a larger cardinality than $|\mathbb{R}|$. [7]

Q.3 (a) Find the power set of each of these sets:

[7]

i) $\{a, b\}$

ii) $\{\phi, \{\phi\}\}$

Definition: The power set $P(S)$ is the set of all subsets of S , including the empty set $\emptyset$ and S itself. If $|S| = n$, then $|P(S)| = 2^n$.

i) Find the power set of $\{a, b\}$

- **Elements:** a, b .
- **Size:** 2 elements. Expected subsets: $2^2 = 4$.
- **Subsets:**
 1. Size 0: $\emptyset$
 2. Size 1: $\{a\}, \{b\}$
 3. Size 2: $\{a, b\}$
- **Answer:**

$$P(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

ii) Find the power set of $\{\emptyset, \{\emptyset\}\}$

- **Let:** $x = \emptyset$ and $y = \{\emptyset\}$.
- **Set becomes:** $S = \{x, y\}$.
- **Size:** 2 elements. Expected subsets: $2^2 = 4$.
- **Subsets of $\{x, y\}$:** $\emptyset, \{x\}, \{y\}, \{x, y\}$.
- **Substitute back original values:**
 1. $\emptyset$
 2. $\{x\} \rightarrow \{\emptyset\}$
 3. $\{y\} \rightarrow \{\{\emptyset\}\}$
 4. $\{x, y\} \rightarrow \{\emptyset, \{\emptyset\}\}$
- **Answer:**

$$P(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$$

(b) Use Cantor's diagonal argument to prove that the set F of all functions $f : (0, 1) \rightarrow \mathbb{R}$ has a larger cardinality than $|\mathbb{R}|$. [7]

Proof:

1. **Cardinality of the Reals:** Let the cardinality of $\mathbb{R}$ be denoted by c (the continuum). Note that the interval $(0, 1)$ also has cardinality c .
2. **Cardinality of the Function Space:** The set F of all functions from $(0, 1)$ to $\mathbb{R}$ can be denoted as $\mathbb{R}^{(0,1)}$. The cardinality of this set is $|F| = |\mathbb{R}|^{|(0,1)|} = c^c$.
3. **Comparison with Power Set:** Consider the set of characteristic functions $g : (0, 1) \rightarrow \{0, 1\}$. This set is equivalent to the power set of $(0, 1)$, denoted $P((0, 1))$. The cardinality of this subset of functions is 2^c . Clearly, the set of *all* real-valued functions F is at least as large as this subset.

$$|F| \geq 2^c$$

4. **Cantor's Theorem:** Cantor's Theorem states that for any set X , the cardinality of its power set $P(X)$ is strictly greater than the cardinality of X .

$$|P(X)| > |X|$$

Applying this to the set of real numbers (or the interval $(0, 1)$):

$$2^{\mathfrak{c}} > \mathfrak{c}$$

Conclusion: Since $|F| \geq 2^{\mathfrak{c}}$ and $2^{\mathfrak{c}} > \mathfrak{c}$ (where $\mathfrak{c} = |\mathbb{R}|$), it follows that:

$$|F| > |\mathbb{R}|$$

Thus, the set of functions F has a strictly larger cardinality than the set of real numbers $\mathbb{R}$

Q.4 Determine if the sets are countable or uncountable:

- a. The set A of all functions $g : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$.
- b. The set B of all functions $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$.

Problem: Determine if the set A of all functions $g : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$ is countable or uncountable.

Solution: Uncountable

Reasoning:

1. This set A represents the set of all infinite sequences of positive integers: $(a_1, a_2, a_3, \dots)$ where each $a_i \in \mathbb{Z}_+$.
2. Let's compare this to a known uncountable set. Consider the set S of all infinite binary sequences (sequences of 0s and 1s). The set S corresponds to functions $h : \mathbb{Z}_+ \rightarrow \{0, 1\}$.
3. Cantor's Diagonal Argument proves that S is uncountable. (If you assume listability, you can flip the diagonal bit to create a sequence not in the list).
4. The set A (functions to $\mathbb{Z}_+$) contains S as a subset (since $\{0, 1\} \subset \mathbb{Z}_+$).
5. Since a subset of A is uncountable, A itself must be uncountable.
6. Formally, $|A| = |\mathbb{Z}_+|^{\mathbb{Z}_+} = \aleph_0^{\aleph_0}$. We know $2^{\aleph_0} \leq \aleph_0^{\aleph_0}$. Since $2^{\aleph_0}$ is uncountable (cardinality of the continuum c), A is uncountable.

Q.4 (b) Countability of Functions to $\{0, 1\}$

Problem: Determine if the set B of all functions $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$ is countable or uncountable.

Solution: Uncountable

Reasoning:

1. **Equivalence to Power Set:** Any function $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$ can be identified with a subset of $\mathbb{Z}_+$. For a given function f , we construct the subset $X_f = \{n \in \mathbb{Z}_+ \mid f(n) = 1\}$. Conversely, for any subset $X \subseteq \mathbb{Z}_+$, we can define a characteristic function. Thus, the set B is bijectively equivalent to the Power Set of $\mathbb{Z}_+$, denoted $P(\mathbb{Z}_+)$.

2. **Cantor's Theorem:** Cantor's Theorem states that for any set S , $|P(S)| > |S|$. Here, $S = \mathbb{Z}_+$, which is countably infinite ($\aleph_0$).

3. **Cardinality:** The cardinality of B is $|P(\mathbb{Z}_+)| = 2^{\aleph_0}$. It is a well-known result that $2^{\aleph_0} = \mathfrak{c}$ (the cardinality of the real numbers), which is uncountable.

4. **Conclusion:** The set B is uncountable.

Q.5 Prove the following by using the principle of mathematical induction for all $n \in \mathbb{N}$: [14]

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Proof:

Step 1: Base Case ($n = 1$)

- **LHS:** $1^3 = 1$
- **RHS:** $\left(\frac{1(1+1)}{2} \right)^2 = \left(\frac{2}{2} \right)^2 = 1^2 = 1$
- LHS = RHS. The statement is true for $n = 1$.

Step 2: Inductive Hypothesis Assume the statement is true for some integer $k \geq 1$:

$$P(k) : 1^3 + 2^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2$$

Step 3: Inductive Step We need to prove that $P(k + 1)$ is true, i.e.,

$$1^3 + \dots + k^3 + (k + 1)^3 = \left(\frac{(k + 1)(k + 2)}{2} \right)^2$$

Start with the LHS of $P(k + 1)$:

$$\text{LHS} = [1^3 + \dots + k^3] + (k + 1)^3$$

Substitute the Inductive Hypothesis $P(k)$:

$$\begin{aligned} &= \left(\frac{k(k + 1)}{2} \right)^2 + (k + 1)^3 \\ &= \frac{k^2(k + 1)^2}{4} + (k + 1)^3 \end{aligned}$$

Factor out $(k + 1)^2$:

$$= (k + 1)^2 \left[\frac{k^2}{4} + (k + 1) \right]$$

Take common denominator:

$$\begin{aligned} &= (k+1)^2 \left[\frac{k^2 + 4(k+1)}{4} \right] \\ &= (k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right] \end{aligned}$$

Recognize that $k^2 + 4k + 4 = (k+2)^2$:

$$\begin{aligned} &= (k+1)^2 \frac{(k+2)^2}{4} \\ &= \left(\frac{(k+1)(k+2)}{2} \right)^2 \end{aligned}$$

This equals the RHS of $P(k+1)$.

Conclusion: By the principle of mathematical induction, the formula holds for all $n \in \mathbb{N}$.

Q.6 State and prove the Division Algorithm theorem using the well-ordering principle. [14]

Statement: Given integers a and b with $b > 0$, there exist unique integers q and r such that $a = bq + r$ and $0 \leq r < b$.

Proof using Well-Ordering Principle:

Part 1: Existence Consider the set S of all non-negative integers of the form $a - bx$:

$$S = \{a - bx \mid x \in \mathbb{Z} \text{ and } a - bx \geq 0\}$$

1. **Show S is non-empty:**

- If $a \geq 0$, choose $x = 0$, then $a \in S$.
- If $a < 0$, choose $x = a$. Since $b \geq 1$, $a - ba = a(1 - b)$. Since a is negative and $(1 - b) \leq 0$, the product is ≥ 0 . So $a - ba \in S$.

2. **Apply Well-Ordering Principle:** The Well-Ordering Principle states that every non-empty set of non-negative integers contains a least element. Let r be the least element of S .

3. **Formulate q and r :** Since $r \in S$, there exists an integer q such that $r = a - bq$. Rearranging gives $a = bq + r$. Also, by definition of S , $r \geq 0$.

4. **Prove $r < b$:** We prove by contradiction. Assume $r \geq b$. Then $r - b \geq 0$. Substitute r : $r - b = (a - bq) - b = a - b(q + 1)$. This implies that $a - b(q + 1)$ is a non-negative integer of the correct form, so $(r - b) \in S$. But $r - b < r$ (since $b > 0$). This contradicts the fact that r is the *least* element of S . Therefore, assumption is false, and $r < b$.

Part 2: Uniqueness Suppose there are two distinct pairs (q, r) and (q', r') :

$$a = bq + r = bq' + r'$$

Assume $q \neq q'$. Rearrange:

$$b(q - q') = r' - r$$

Taking absolute values:

$$b|q - q'| = |r' - r|$$

Since $q \neq q'$, $|q - q'| \geq 1$, so LHS $\geq b$. However, since $0 \leq r, r' < b$, the difference $|r' - r|$ must be strictly less than b .

$$b \leq |r' - r| < b$$

This is a contradiction ($b < b$ is impossible). Therefore, $q = q'$, which implies $r = r'$.

(b) A grocery store employee is stocking apples. Each apple is a different color. There are 10 apples left in the box and the employee pulls out 2 of them at random. What is the probability that the employee pulls out one pink apple and one yellow apple? [7]

Problem: A box has 10 apples, each a different color. Two apples are pulled at random. Find the probability that one is **pink** and one is **yellow**.

Solution:

1. **Total number of ways to pick 2 apples:** Since order does not matter (we just want a pair), we use combinations.

$$\text{Total Outcomes} = \binom{10}{2} = \frac{10 \times 9}{2 \times 1} = 45$$

2. **Identify Favorable Outcomes:** We need the pair to contain exactly the Pink apple and the Yellow apple.

- Ways to choose the Pink apple = $\binom{1}{1} = 1$.
- Ways to choose the Yellow apple = $\binom{1}{1} = 1$.
- Ways to choose 0 other apples = $\binom{8}{0} = 1$.

Total favorable pairs = $1 \times 1 = 1$ pair (The pair {Pink, Yellow}).

3. Calculate Probability:

$$P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

$$P(\text{Pink and Yellow}) = \frac{1}{45}$$

Alternative Method (Sequential): P(1st is Pink AND 2nd is Yellow) OR P(1st is Yellow AND 2nd is Pink).

$$\begin{aligned} &= \left(\frac{1}{10} \times \frac{1}{9} \right) + \left(\frac{1}{10} \times \frac{1}{9} \right) \\ &= \frac{1}{90} + \frac{1}{90} = \frac{2}{90} = \frac{1}{45} \end{aligned}$$

(a) Given $A = \{1, 2, 3\}$, $B = \{7, 8\}$ and $R = \{(1, 7), (2, 7), (1, 8), (3, 8)\}$, find R^{-1} (inverse of R) and R' (complement of R). [7]

Problem: Given $A = \{1, 2, 3\}$, $B = \{7, 8\}$ and $R = \{(1, 7), (2, 7), (1, 8), (3, 8)\}$. Find R^{-1} and R' .

i) Inverse Relation R^{-1}

Definition: $R^{-1} = \{(b, a) \mid (a, b) \in R\}$. Simply swap the coordinates of every pair in R .

- $(1, 7) \rightarrow (7, 1)$
- $(2, 7) \rightarrow (7, 2)$
- $(1, 8) \rightarrow (8, 1)$
- $(3, 8) \rightarrow (8, 3)$

Answer:

$$R^{-1} = \{(7, 1), (7, 2), (8, 1), (8, 3)\}$$

ii) Complement Relation R'

Definition: $R' = (A \times B) - R$. It contains all pairs in the Cartesian product that are NOT in R .

Step 1: Find Cartesian Product $A \times B$ $A = \{1, 2, 3\}$, $B = \{7, 8\}$. Size = $3 \times 2 = 6$ pairs.

$$A \times B = \{(1, 7), (1, 8), (2, 7), (2, 8), (3, 7), (3, 8)\}$$

Step 2: Remove elements present in R R contains: $(1, 7), (1, 8), (2, 7), (3, 8)$.

Remaining pairs:

- $(2, 8)$
- $(3, 7)$

Answer:

$$R' = \{(2, 8), (3, 7)\}$$

(b) Consider $A = B = C = \mathbb{R}$ (set of real numbers) and let $f : A \rightarrow B$, $g : B \rightarrow C$ defined by $f(x) = x + 9$ and $g(y) = y^2 + 3$. Find the following composition functions: [7]

$$(f \circ g)(a), \quad (g \circ f)(a), \quad (g \circ f)(3), \quad (f \circ g)(-3)$$

Problem: Let $f(x) = x + 9$ and $g(y) = y^2 + 3$. Find the specific values.

1. $(f \circ g)(a)$

$$(f \circ g)(a) = f(g(a))$$

Substitute $g(a) = a^2 + 3$:

$$f(a^2 + 3) = (a^2 + 3) + 9 = a^2 + 12$$

2. $(g \circ f)(a)$

$$(g \circ f)(a) = g(f(a))$$

Substitute $f(a) = a + 9$:

$$g(a + 9) = (a + 9)^2 + 3$$

Expand:

$$(a^2 + 18a + 81) + 3 = a^2 + 18a + 84$$

3. $(g \circ f)(3)$ We can use the formula derived in part 2, setting $a = 3$:

$$(3)^2 + 18(3) + 84$$

$$9 + 54 + 84$$

$$63 + 84 = 147$$

Check manually: $f(3) = 3 + 9 = 12$. $g(12) = 12^2 + 3 = 144 + 3 = 147$. Correct.

4. $(f \circ g)(-3)$ We can use the formula derived in part 1, setting $a = -3$:

$$(-3)^2 + 12$$

$$9 + 12 = 21$$

Check manually: $g(-3) = (-3)^2 + 3 = 9 + 3 = 12$. $f(12) = 12 + 9 = 21$. Correct.

(b) Use the Euclidean algorithm to find the greatest common divisor of each pair of integers: [7]

(i) 60, 90

(ii) 414, 662

Problem: Use Euclidean algorithm to find the GCD.

(i) GCD of 60, 90

Divide larger by smaller:

$$90 = 1 \times 60 + 30$$

Now divide previous divisor (60) by remainder (30):

$$60 = 2 \times 30 + 0$$

Since remainder is 0, the last non-zero remainder is the GCD. **GCD = 30.**

(ii) GCD of 414, 662

Step 1: $662 \div 414$

$$662 = 1 \times 414 + 248$$

Step 2: $414 \div 248$

$$414 = 1 \times 248 + 166$$

Step 3: $248 \div 166$

$$248 = 1 \times 166 + 82$$

Step 4: $166 \div 82$

$$166 = 2 \times 82 + 2$$

Step 5: $82 \div 2$

$$82 = 41 \times 2 + 0$$

Remainder is 0. The last non-zero remainder was 2. **GCD = 2.**

(a) What is the negation of each of the following propositions?

- (i) Today is Tuesday.
- (ii) A cow is an animal.
- (iii) No one wants to buy my house.

Problem: Find the negation of the following propositions.

(i) "Today is Tuesday."

- **Negation:** Today is **not** Tuesday.

(ii) "A cow is an animal."

- **Negation:** A cow is **not** an animal.

(iii) "No one wants to buy my house."

- **Logical Form:** $\forall x(\neg \text{WantsToBuy}(x, \text{House}))$.
- **Rule:** The negation of "For all... not..." is "There exists...". $\neg(\forall x, \neg P(x)) \equiv \exists x, P(x)$.
- **Negation:** **Someone** wants to buy my house. (Or: At least one person wants to buy my house).

(b) Determine the truth value of each of the following statements:

(i) $4 + 3 = 7$ and $6 + 2 = 8$

(ii) $2 + 3 = 4$ and $3 + 1 = 2$

Problem: Determine the truth value.

(i) $4 + 3 = 7$ **and** $6 + 2 = 8$

- Statement A: $4 + 3 = 7$. This is **True**.
- Statement B: $6 + 2 = 8$. This is **True**.
- Compound: True **AND** True.
- **Result: TRUE.**

(ii) $2 + 3 = 4$ **and** $3 + 1 = 2$

- Statement A: $2 + 3 = 4$. This is **False** (it equals 5).
- Statement B: $3 + 1 = 2$. This is **False** (it equals 4).
- Compound: False **AND** False.
- **Result: FALSE.**

(Note: Even if one was True, "False AND ..." is always False.)

Show: $p \iff q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

Proof using Truth Table:

We construct a truth table for both expressions.

p	q	LHS: $p \iff q$	$p \rightarrow q$	$q \rightarrow p$	RHS: $(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

Conclusion: Compare the column for **LHS** ($p \iff q$) and **RHS** ($(p \rightarrow q) \wedge (q \rightarrow p)$).

- Row 1: T vs T
- Row 2: F vs F
- Row 3: F vs F
- Row 4: T vs T

Since the truth values match in every possible case, the two logical expressions are equivalent.